ANIRUDH SARVE

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PROFESSIONAL SUMMARY

Computer Engineering undergraduate specializing in Applied Machine Learning, Computer Vision, Sensor Fusion, and Distributed Systems. Proven track record in formulating physics-constrained Kalman filters, cross-modal remote sensing inpainting networks, and edge runtimes (ONNX, WebAssembly, ARM Cortex-A) delivering mathematically bounded precision, sub-3 ms latency budgets, and robust real-world performance.

EDUCATION

Vidyalankar Institute of Technology (VIT), Mumbai — **B.E. in Computer Engineering** **Expected May 2028**

Cumulative GPA: 8.90 / 10.0 | Affiliated with University of Mumbai | **Class XII (HSC): 92.60%** (Distinction)

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript (Node.js), SQL, Go (Basics), HTML5/CSS3.
AI & Machine Learning: PyTorch, TensorFlow, ONNX, OpenCV, Faster-Whisper, Scikit-learn, 15-State ESKF, Sensor Fusion, ResU-Net.
Web & Systems: Next.js 16, React 19, FastAPI, Express.js, WebAssembly (WASM), WebGL, WebSockets, Docker, Linux, Git.
Databases & Cloud: PostgreSQL, MongoDB, Upstash Redis (Distributed Locks), Qdrant Vector DB, SQLite (FTS5), AWS, GCP.

PROFESSIONAL EXPERIENCE

NxtChaptr — **Software Developer (Edge AI & Full-Stack Infrastructure)** **Nov 2024 – Aug 2025**

- Architected client-side edge computer vision using U²-Net via **WebAssembly/WebGL**, processing image matting in **<3.5s** and eliminating server GPU fees.
- Engineered automated catalog metadata extraction microservices using **Google Gemini Vision**, reducing listing creation latency from 8 min to **<30s**.
- Integrated Upstash Redis atomic locks (SET NX EX) and Ably WebSockets to eliminate race-condition overselling on concurrent 1-of-1 product drops.

Wroomie — **Software Developer (Backend & Applied ML)** **Nov 2023 – Jan 2024**

- Developed multi-attribute compatibility scoring clustering user preferences across 4 behavioral vectors with **94% match satisfaction**.
- Optimized MongoDB geospatial queries with 2dsphere indexing, decreasing room discovery query latency by **35%** across university zones.

FEATURED TECHNICAL PROJECTS

Neural-Inertial Dead-Reckoning Engine with 15-State Error-State Kalman Filter (ISRO SIH)
Technologies: PyTorch, 15-State ESKF, 5-Head Temporal Attention GRU, ONNX, C++, ARM Cortex-A

- Engineered hybrid dead-reckoning engine holding vehicle position drift to **≤ 1.85 m** across 120s GPS outages (a **98.4% drift reduction** vs standard EKF).
- Fused 10 Hz raw IMU with 15-state ESKF, enforcing physical Non-Holonomic Constraints (NHC) and ZUPT updates at sub-3 ms CPU latency on ARM Cortex-A ECUs.

Multimodal Optical-SAR Fusion ResU-Net with Cross-Modal Attention (ISRO LISS-IV)
Technologies: PyTorch, ResU-Net, Cross-Modal Attention, Sentinel-1 SAR (C-Band), GeoTIFF, GDAL

- Constructed generative AI framework for automated cloud removal in ISRO Resourcesat-2/2A LISS-IV satellite imagery (5.0 m resolution), fusing optical and SAR radar.
- Formulated differentiable Spectral Angle Mapper (SAM) loss, reaching **32.09 dB PSNR**, **0.7167 SSIM**, and tight **6.17° SAM angle** on validation scenes.

Neuro-Symbolic Hybrid Retrieval Engine & Standards Compliance Platform (SIH 26108)
Technologies: React 18, TypeScript, FastAPI, SQLite FTS5 BM25, ONNX Embeddings, PaddleOCR

- Built 3-way hybrid retrieval engine (SQLite FTS5 BM25 + regex + dense embeddings) via Reciprocal Rank Fusion across 19,423 BIS engineering standards.
- Integrated verification kernel with PaddleOCR pipeline achieving **99.28% OCR confidence** on complex tender BoQ tables.

KEY ACHIEVEMENTS & AWARDS

- **Winner & Lead Architect, ISRO Smart India Hackathon (SIH 2026):** Authored winning AI-ESKF sensor-fusion architecture for PS 26168.
- **Finalist, Indian Rover Challenge (IRC):** Designed onboard kinematic guidance telemetry and dead-reckoning estimators for Mars rover prototype.
- **Scholastic Honors:** 8.90 / 10.0 CGPA at VIT Mumbai; 92.60% (Distinction) in Maharashtra State Board HSC Examination.
- **Open Source Contributor:** Published vector PDF streaming compilers and Model Context Protocol (MCP) tool integration servers on GitHub.

HOBBIES & PERSONAL INTERESTS

Technical & Creative Pursuits: Embedded robotics tinkering (ROS2 / micro-ROS), Autonomous drone navigation, Competitive chess (Tactical positional play), Amateur astronomy & satellite orbit tracking, Technical writing & systems architecture blogging.